

Syed Mosaddik Hossain Ifty

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Robotics & Intelligent Systems | Embedded & Autonomous Systems | Computer Vision
[Research Portfolio](#) | [Google Scholar](#) | [EPSRG](#) | [LinkedIn](#)

◆ EDUCATION

Bachelor of Science in Electrical and Electronic Engineering (2019 - 2025)

BRAC University, Dhaka, Bangladesh

Major: Robotics and Intelligent System | **CGPA:** 3.06/4.00

Core Coursework: Advanced Mathematics | Embedded Systems | Artificial Intelligence | Signal Processing | Control Systems

◆ RESEARCH FOCUS

Electrical and Electronic Engineering graduate and Research Assistant working at the intersection of embedded control, computer vision, UAV-assisted inspection, mechatronics, and photovoltaic automation. Current work integrates STM32-based control with computer-vision pipelines for intelligent solar-panel thermal inspection and robotic cleaning systems.

◆ RESEARCH EXPERIENCE

Research Assistant, EPSRG, Department of Electrical and Electronic Engineering, BRAC University | Nov 2025 - Oct 2026
| Dhaka, Bangladesh

- Wrote project funding proposals that secured **USD 17,000** for industry-grade development of Solar Panel Inspection and Cleaning Systems (SPICS)
- Continued funded R&D of SPICS, developing mechanically guided solar-cleaning robots and C/C++ control logic on STM32 microcontrollers for solar power plants in Bangladesh.
- Integrated DC and Servo motor actuation with closed-loop feedback for reliable mechanical guidance and cleaning operation.
- Developed Python-based computer-vision pipelines and fine-tuned YOLO models for photovoltaic hotspot and environmental-degradation detection.

Advisor: Prof. Md. Mosaddequr Rahman, PhD | Professor and Chairperson | Department of Electrical and Electronic Engineering | BRAC University

Current Research Direction: Embedded control for photovoltaic cleaning, UAV-based intelligent photovoltaic inspection, autonomous maintenance, and deployment-oriented robotic systems.

Research Assistant, Department of Electrical and Electronic Engineering, BRAC University | Apr 2025 - Aug 2025

- Contributed to a BIGD- and H&M Foundation-funded pilot study on agrivoltaics and sustainable business models in Bangladesh, supporting research on the integration of photovoltaic generation with agricultural applications.
- Predicted energy generation for the project site in PVSyst

◆ SELECTED PUBLICATIONS | Citation: 133 | h-index: 4 | i10-index: 4

Research Outputs: 1 accepted first-author IEEE conference paper | 3 first-authored peer-reviewed review articles

- **S. M. H. Ifty**, S. I. Alif, M. Rahman, and M. Islam. "Design and Assessment of a Solar Power Plant Inspection and Cleaning Systems". *TENCON 2026 - 2026 IEEE Region 10 Conference (TENCON)*, Bali, Indonesia, October 10–13, 2026. (Accepted for presentation)
- **S. M. H. Ifty**, F. Irin, M. S. S. Shovon, M. H. H. Amjad, P. K. Bhowmik, R. Ahmed, M. R. Ashakin, B. Hossain, M. Mushfiq, A. Sattar, R. Chowdhury, and A. R. Sunny, "Advancements, applications, and future directions of artificial intelligence in healthcare," *Journal of Angiotherapy*, vol. 8, no. 8, pp. 1–18, Aug. 2024. [DOI](#)
- **S. M. H. Ifty**, M. R. Ashakin, B. Hossain, S. Afrin, A. Sattar, R. Chowdhury, M. I. Tusher, P. K. Bhowmik, M. T. Mia, T. Islam, M. Tufael, and A. R. Sunny, "IoT-based smart agriculture in Bangladesh: An overview," *Applied Agriculture Sciences*, vol. 1, no. 1, pp. 1–10, Feb. 2023. [DOI](#)

◆ RESEARCH INTERESTS AND GRADUATE DIRECTION

Research Interests: Autonomous and field robotics | Computer vision | Embedded and edge intelligence | Intelligent maintenance systems

Interested in research that integrates visual perception, embedded control, and physical actuation for robust autonomous robotic and mechatronic systems in real-world environments.

◆ FLAGSHIP PROJECT

Project Team Member

Solar Panel Monitoring and Cleaning Systems | EEE400 Final Year Design Project | Sep 2023 - Oct 2024

Project Scope: UAV inspection | clean-vs-dusty CNN classification | ESP32-based cleaning rover | IoT ground-station workflow | embedded control | physical test-bed evaluation

- Contributed to a four-member team that developed an integrated system combining UAV-based photovoltaic panel inspection, deep-learning-based assessment, IoT communication, and semi-autonomous rover cleaning.
- Built an ESP32-based cleaning architecture with DC motors, proximity and limit sensing, SolidWorks/CAD design, and Proteus circuit simulation.
- Developed a UAV inspection pipeline using Python and compared Custom CNN, ResNet50, and VGG16 models; used ArduPilot and Mission Planner for UAV-tuning and mission control.

Evaluation: Custom CNN achieved 79.0% classification accuracy and 90.1% precision for the Dusty class; the rover cleaned 315×20 W test panels per battery cycle, while the UAV inspected 75 panels per battery cycle.

Recognition: Best Project Award, BSRM School of Engineering, BRAC University, selected among 10 Summer 2024 graduating FYDP groups.

Research Continuation: The project later became the foundation for continued SPICS R&D involving STM32-based robotic control and UAV-based photovoltaic inspection.

◆ ADDITIONAL ENGINEERING PROJECTS

Signboard Detection in Dhaka Streets Using Deep Learning | EEE476 Image Processing | Jan - May 2024

- Curated and annotated a novel street-image dataset in CVAT, trained YOLO-based object-detection model in Python and Google Colab, and evaluated performance using Precision, Recall, IoU, and mAP.
- Aimed at aiding in future autonomous vehicle operation in Dhaka city.

Automated Four-Way Traffic Intersection Management System | EEE365 Microprocessors and Interfacing | Jan - May 2023

- Programmed an ATmega32 microcontroller with Embedded C in CodeVisionAVR to coordinate synced servo-controlled barricades and LED signal logic for a multi-lane intersection.

Temperature Control Box | EEE305 Control Systems | Jun - Sep 2023

- Built and simulated a closed-loop thermal regulation system using LM35 sensing, Arduino Uno controller, ADC, interrupt handling, relay actuation, and Proteus validation.

◆ TEACHING AND EARLIER RESEARCH EXPERIENCE

Programming and Robotics Instructor, STEMON Bangladesh | Jan 2025 - Present | Weekend Program

- Teaching Scratch and Python programming, and introductory Arduino-based sensor and motor-control integration to students in Grades 5–12.

Undergraduate Research Assistant, Pathfinder Research and Consultancy Center | Oct 2022 - Aug 2024 | Remote

- Worked on IoT for smart agriculture and artificial intelligence in healthcare; this research culminated in two first-authored peer-reviewed review publications.

◆ TECHNICAL AND RESEARCH SKILLS

Programming & Scientific Computing: Python | C/C++ | Embedded C | MATLAB

Deep Learning & Computer Vision: YOLO | Custom CNNs | VGG16 | ResNet50 | CVAT | OpenCV | Image processing and analysis

Embedded Systems & Control: STM32 | ESP32 | Arduino Uno | ATmega32 | ADC | Interrupts | Sensor interfacing | Actuator control

Robotics & UAV Systems: ArduPilot | Mission Planner | Mechatronic integration | DC/BLDC/Servo motors | Feedback control

Research: Literature review | Dataset preparation & annotation | Model evaluation | Technical writing | Research proposal development

◆ LANGUAGE SKILLS

Bengali: Native

English: IELTS Academic - Overall Band 7.0 (CEFR: C1)

- Listening C1 | Reading C1 | Speaking B2 | Writing B2